Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/06/2005
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Project Title:	Novel Optical Diagnostic Techniques for Studying Particle Contact and Deposition Upon a Large Cylinder in a Sheared Suspension		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
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Space Biology Element:	None		
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Task Description:	Dr. Yoda was supported by the NASA Office of Biological and Physical Research on a four-year grant from March 2000 through November 2004 for a ground-based study on the fluid dynamics of sheared particle-liquid suspensions and the impact of differential particle-fluid inertia on such flows. Such inertial effects can only be observed in reduced-gravity environments since they are overwhelmed by buoyancy effects on Earth. Moreover, these inertial effects will have a significant impact upon suspension flows in microgravity. Suspension dynamics are of importance in a wide variety of advanced life systems applications, including water reclamation and dust mitigation in confined habitats. The objectives of this research project were: 1) To study the fluid dynamics of sheared particle-liquid suspensions and the impact of differential particle-fluid inertia 2) To develop new techniques for observing suspension particle contact and deposition upon solid surfaces. This study establishes the experimental methodology and techniques		

required to carry out micro- or reduced-gravity experiments on the impact of differential particle-fluid inertia on particle-fluid interactions and suspension flows. These effects can only be observed in reduced-gravity environments since they are overwhelmed by buoyancy effects on Earth. Moreover, these inertial effects will have a significant impact upon suspension flows in microgravity. Suspension dynamics are of importance in a wide variety of advanced life systems applications, including water reclamation and dust mitigation in confined habitats.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The dynamics of neutrally buoyant prolate and oblate spheroids immersed in a simply sheared liquid were studied using a plane Couette channel as a simple three-dimensional model of suspended particle dynamics in a dilute suspension (i.e., in the absence of particle-particle interactions). Recent two-dimensional numerical studies using the lattice-Boltzmann technique have suggested that these bodies, when constrained to rotate about one of their principal axes, undergo a rotation transition from a time-periodic to a stationary state above a critical shear-based Reynolds number and that this transition is independent of particle shape except for a sphere in simple shear flow [Ding & Aidun 2000]. More recent three-dimensional lattice-Boltzmann simulations suggest, however, that the dynamics of neutrally buoyant spheroidal particles in (plane) Couette flow—when free to rotate about any axis—are more complex, with prolate spheroids undergoing two rotation transitions and nutation between these transitions, while oblate spheroids undergo only one rotation transition [Qi and Luo 2003]. These new results demonstrate that the classical problem of a sphere in simple shear flow is actually a special case that cannot be generalized to other particle shapes, and that even slight deviations from sphericity due to surface roughness (for example) may have a significant impact on suspension particle dynamics. A plane Couette flow facility based upon a previous design [Zettner & Yoda 2001], was designed and built to study cylinders and spheroids in simply sheared single-phase liquids and suspension flows. The linearity and steadiness of the simple shear flow the test section were characterized using both single-component laser-Doppler velocimetry (LDV) and particle-image velocimetry (PIV). For conditions corresponding to a particle- and shear-based Reynolds number $Re_G < 20$ (based upon a, the particle or spheroid semimajor dimension and G, the shear rate), the linearity of the flow, quantified by the deviation between the average measured and the expected streamwise velocity components normalized by the belt speed, was less than 6%.

For the three-dimensional studies, spheroids were manufactured from three different materials for these studies:

1) DMS Somos 7545 resin (specific gravity SG = 1.20) was fabricated into spheroids using stereolithography rapid prototyping; 2) Polydimethylsiloxane (PDMS) (SG = 1.07) was molded into spheroids using molds manufactured with stereolithography rapid prototyping; 3) Acrylic (polymethylmethacrylate, or PMMA) (SG = 1.17) was milled into spheroids using a micro-CNC (computerized numerical control) mill.

In all cases, the spheroids had an aspect ratio AR = b/a = 0.33 and 0.5 (b is the seminor axis of the spheroid), and a = 0.45 cm, giving a confinement ratio in the test section a/h = 0.35. A variety of spheroids with different SG were developed so measurements could be carried out for both single-phase and suspension flows.

Rotation periods were measured for spheroids constrained to rotate about both their major and minor axes. No rotation transition was observed for both types of prolate spheroids rotating about their major axes, presumably because the transition (if it occurs) occurs at a critical Reynolds number Re_{cr} beyond the range attainable with the plane Couette channel (i.e., $Re_{cr} > 35$). A transition to a stationary state was observed, however, for both prolate spheroids rotating about their minor axes, with $Re_{cr} = 19.2$ and 33.5 for AR = 0.33 and 0.5, respectively.

No rotation transition was observed for the oblate spheroids rotating about their minor axes, again most likely because the transition (if it occurs) occurs at a critical Reynolds number Re_{cr} beyond the range attainable in these experiments. A transition to a stationary state was observed; however, for both oblate spheroids rotating about their major axes, with $Re_{cr} = 10.3$ and 30.5 for AR = 0.33 and 0.5, respectively.

The prolate and oblate spheroid cases that do show a transition closely follow the scaling law proposed by Ding and Aidun (2003) as Re_G approaches Re_{cr}. Most interesting, although these bodies were constrained to rotate about one of their principal axes, we observed this transition at $Re_{cr} = 33.5$ for an AR = 0.5 prolate spheroid rotating about its minor axis, in direct contradiction of the lattice-Boltzmann results reported by Qi and Luo (2003), who reported that this body would remain rotating about its minor axis for Re_G less than about 50.

References

E.-J. Ding and C. K. Aidun (2000) The dynamics and scaling law for particles suspended in shear flow with inertia. J. Fluid Mech. 423, 317

D. Qi and L.-S. Luo (2003) Rotational and orientational behaviour of three-dimensional spheroidal particles in Couette flow. J. Fluid Mech. 477, 201

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